CleanLoop: Project Brief

Team Git Happens · Environmental Hacks (Bharat Builds Tour, Event 02) · Track 03: Waste and Energy · Build window Oct 8 to 11, 2026

1. Problem

- Mixed waste at the source makes recycling nearly impossible. Most households don't know which bin an item belongs in.
- Collection is reactive. Bins overflow, residents have no idea when the truck will arrive, and complaints go nowhere visible.
- Trucks follow fixed routes whether a lane needs a pickup or not, which wastes diesel and adds CO2.
- Informal recyclers (kabadiwalas) already handle a large share of recyclable and e-waste, but they are invisible to digital systems and hard for residents to find.

Focus area for the demo: one ward or campus in Delhi (for example the DTU area), seeded with real lanes and stops.

2. Solution overview

One mobile-first web app with four roles:

- **Resident:** snaps an item to learn the right bin, sees the truck ETA for their lane, reports overflowing bins, requests recycler or e-waste pickups.
- **Driver:** sees the day's stop list in order and taps Done on each stop.
- **Officer:** gets severity-ranked bin reports, a map, a next-day overflow risk list, and a resolve button with an after-photo.
- **Recycler:** registers, accepts pickup requests and marks them done.

3. The loop (how the pieces connect)

1. **Sort:** a resident photographs an item. Bedrock returns wet, dry, e-waste or hazardous, plus the bin and a disposal tip. Recyclables and e-waste show a Request pickup button.
2. **Collect:** the truck ETA, driver Done button and SNS alerts keep residents informed. A skipped stop alerts the lane and invites a report.
3. **Report:** a photo of an overflowing bin or dump goes to S3, Bedrock scores severity, and SES emails the ward officer with photo, location and map link.

4. **Predict and save:** every night a risk score per lane reorders tomorrow's route. A dashboard shows kilometres, diesel and CO2 saved versus the fixed route.
5. **Recycle:** pickup requests go to the nearest registered recycler, who accepts and completes them.

4. Architecture (AWS)

Layer	Services
Frontend	Amplify Hosting (mobile-first PWA), Cognito (roles: resident, driver, officer, recycler)
API and compute	API Gateway, Lambda
AI	Bedrock multimodal model for sorting and severity (Rekognition as fallback)
Data	DynamoDB (with Streams), S3 for photos
Alerts	SNS for residents (SMS), SES for officers and recyclers
Scheduling	EventBridge (nightly risk job)
Map and routing	Amazon Location Service (maps, route calculator)
Weather input	Open-Meteo forecast API (free, non-commercial, attribution required)
Monitoring	CloudWatch

Key flows

- **Sort:** browser uploads photo to S3 via presigned URL, triggers Lambda, Lambda calls Bedrock with a structured JSON prompt, result returned to the UI.
- **Done tap:** driver app calls API, Lambda marks the stop done and recalculates ETAs, a DynamoDB Stream triggers a Lambda that publishes to the lane's SNS topic.
- **Report:** same upload path, Lambda scores severity, looks up the ward from the location, writes the report and sends the SES email.
- **Nightly job:** EventBridge triggers Lambda, which computes risk scores, reorders the next day's stops and writes the savings estimate.

5. Data model (DynamoDB)

- **Wards:** wardId, name, officerEmail, boundary or centre lat/long
- **Routes:** routeId, wardId, vehicleType (diesel or EV), startTime
- **Stops:** routeId, sequence, lane, lat, long, avgMinutesToNext, status (pending, done, skipped), doneAt
- **Reports:** reportId, wardId, lane, photoKey, severity, confidence, status, createdAt, resolvedAt, afterPhotoKey, lat, long
- **PickupRequests:** requestId, residentId, itemType (dry or e-waste), photoKey, lat, long, status (open, accepted, done), recyclerId
- **Recyclers:** recyclerId, name, phone, area, lat, long, acceptedMaterials, approxRates, consentGiven
- **RiskScores:** lane, date, score, level (low, medium, high), factors
- **Subscriptions:** lane, phone or topic ARN

6. Core logic

ETA for a lane ETA(stop k) = time of the last completed stop + sum of avgMinutesToNext for the stops between it and stop k. Recalculated on every Done tap. The demo uses driver taps and a simulated route, not live GPS.

Overflow risk score (0 to 100, per lane per day) Weighted sum, starting point:

- 30% past overflow reports on the same weekday
- 25% days since the last completed pickup
- 20% missed or skipped stops in the last 7 days
- 15% rain forecast (wet waste gets heavier, bins fill faster)
- 10% market, festival or event day flag

Levels: under 35 low, 35 to 65 medium, above 65 high. Tune weights during the backtest. If time allows, replace with a small SageMaker model.

Route re-ordering and savings

- Visit high-risk lanes first, skip or defer lanes predicted nearly empty.
- Compute both routes with the Location Service route calculator.
- Savings = baseline km minus optimised km. Diesel = km divided by an assumed truck efficiency (state the assumption, for example 3 km per litre). CO2 = litres times 2.68 kg per litre of diesel.
- Label all fuel and CO2 numbers as estimates.

Bedrock outputs (structured JSON)

- Sort: category (wet, dry, e-waste, hazardous), bin, tip, confidence
- Severity: isBinOrDump, severity (low, medium, high), reason, confidence
- If confidence is low, ask for a retake or a manual choice rather than guessing.

7. User interface: features by user

One web app (mobile-first), one login, and the screens change by role.

Resident (phone)

- **Home:** lane selector, truck ETA card ("3 stops away, about 25 min"), and a big Snap button.
- **Snap result:** category (wet, dry, e-waste, hazardous), which bin, a one-line tip, confidence. Dry and e-waste show a Request pickup button.
- **Report a bin:** photo with auto-location, AI severity shown, then a status tracker (Sent, Seen, Resolved) with the officer's after-photo.
- **My pickups:** request status with the recycler's name and phone.
- **Alerts:** SMS when the truck is 2 stops away or your lane was missed.

Driver (phone, big buttons)

- **Today's route:** stops in order, current stop highlighted, progress bar.
- **Stop actions:** large Done and Skip buttons. Skip asks for a reason.
- **Map view:** stops as pins with the next one centred.

Officer (laptop)

- **Reports inbox:** sorted by severity, with photo, location, and a Resolve button that takes an after-photo.
- **Map:** pins coloured by severity.
- **Tomorrow's risk list:** lanes shown red, amber or green, with the proposed route order.
- **Savings card:** km, diesel and CO2 saved versus the fixed route.

Recycler (phone)

- **Register:** name, phone, area, materials accepted, approximate rates.
- **Open requests:** nearby pickups with photo and distance, each with Accept.
- **My jobs:** accepted requests with a Mark done button.